# JKSSB Accounts Assistant (Finance) — Full-Length Mock Examination

**Total Marks:** 120
**Time Allowed:** 120 Minutes (2 Hours)
**Negative Marking:** 0.25 marks per wrong answer

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## Section I: General Knowledge with Special Reference to J&K UT (30 Questions | Q1 – Q30)

**1.** Under the Jammu and Kashmir Reorganisation Act, 2019, which Schedule specifies the oaths or affirmations to be taken by the members of the Legislative Assembly and Ministers of the UT of J&K?
A) Second Schedule
B) Third Schedule
C) Fourth Schedule
D) Fifth Schedule

**2.** Section 57 of the Jammu and Kashmir Reorganisation Act, 2019 deals with:
A) High Court of Jammu and Kashmir
B) Abolition of the Jammu and Kashmir Legislative Council
C) Provisions as to Delimitation of Constituencies
D) Distribution of Revenues

**3.** In the context of transport infrastructure in J&K, the world's highest railway pier bridge is being constructed across which river?
A) Chenab
B) Jhelum
C) Ravi
D) Ijai

**4.** Which pass connects the Kashmir Valley with the Kargil and Leh districts of Ladakh?
A) Banihal Pass
B) Zoji La
C) Pir Panjal Pass
D) Sinthan Top

**5.** The historic Treaty of Amritsar (1846), which marked the formal creation of the Princely State of Jammu and Kashmir, was signed between the British East India Company and:

A) Maharaja Ranjit Singh
B) Maharaja Gulab Singh
C) Maharaja Hari Singh
D) Maharaja Pratap Singh

**6.** Who among the following compiled the monumental historical chronicle *Rajatarangini* during the reign of King Jaisimha?
A) Bilhana
B) Kalhana
C) Abhinavagupta
D) Jonaraja

**7.** Which lake in Jammu and Kashmir is designated as a Ramsar Site and is renowned as one of the largest freshwater lakes in Asia, fed by the Jhelum River?
A) Dal Lake
B) Mansar Lake
C) Wular Lake
D) Pangong Tso

**8.** The saffron cultivation in Jammu and Kashmir is predominantly concentrated on the lacustrine plateaus known locally as:
A) Bet
B) Karewas
C) Terai
D) Dhars

**9.** The Martand Sun Temple in Anantnag was commissioned by which ruler of the Karkota dynasty?
A) Durlabhavardhana
B) Lalitaditya Muktapida
C) Avantivarman
D) Didda

**10.** Which national park in Jammu and Kashmir is the primary habitat of the critically endangered Hangul (*Cervus hanglu hanglu*)?
A) Kishtwar National Park
B) Dachigam National Park
C) Kazinag National Park
D) Salim Ali National Park

**11.** The Kishanganga Hydroelectric Project is constructed on the Kishanganga River, which is a major tributary of the:
A) Indus
B) Chenab

C) Jhelum
D) Tawi

**12.** Surinsar and Mansar Lakes, considered twin freshwater bodies with high religious and biological significance, are located in which physiographic division of Jammu and Kashmir?
A) Greater Himalayas
B) Middle Himalayas
C) Shivalik Hills
D) Zanskar Range

**13.** The traditional folk theatre of Kashmir combining satire, mime, music, and social commentary is known as:
A) Bhand Pather
B) Kud Dance
C) Rouf
D) Hafiza

**14.** The Baglihar and Salal hydroelectric power stations are constructed across which of the following rivers?
A) Jhelum
B) Ravi
C) Chenab
D) Sutlej

**15.** According to the 2011 Census of India, which Indian state recorded the lowest sex ratio?
A) Punjab
B) Haryana
C) Rajasthan
D) Bihar

**16.** Which strait connects the Persian Gulf with the Gulf of Oman and is a strategic chokepoint for international petroleum shipments?
A) Strait of Malacca
B) Bab-el-Mandeb
C) Strait of Hormuz
D) Bosphorus Strait

**17.** The Tropic of Cancer (`$23^\circ 30'\text{ N}$`) does NOT pass through which of the following Indian states?
A) Gujarat
B) Odisha
C) Chhattisgarh
D) Tripura

**18.** The historic Non-Cooperation Movement was called off by Mahatma Gandhi following which violent incident in February 1922?
A) Jallianwala Bagh Massacre
B) Kakori Conspiracy
C) Chauri Chaura Incident
D) Chittagong Armoury Raid

**19.** The Grand Anicut (Kallanai), one of the oldest water-regulation structures in the world, was built across which river by Karikala Chola?
A) Krishna
B) Godavari
C) Kaveri
D) Mahanadi

**20.** National Waterway-1 (NW-1) in India operates across which river system?
A) Brahmaputra (Dhubri to Sadiya)
B) Ganga-Bhagirathi-Hooghly (Prayagraj to Haldia)
C) West Coast Canal (Kottapuram to Kollam)
D) Godavari-Krishna stretch

**21.** Which atmospheric layer contains the maximum concentration of ozone that shields the Earth from harmful solar ultraviolet radiations?
A) Troposphere
B) Stratosphere
C) Mesosphere
D) Thermosphere

**22.** The Ramsar Convention, signed in 1971 in Ramsar, Iran, is an intergovernmental treaty primarily focused on the conservation of:
A) Endangered mammals
B) Marine coral reefs
C) Wetlands
D) Mangrove forests exclusively

**23.** Winter rainfall in northwestern India, including Jammu and Kashmir, is caused primarily by:
A) South-West Monsoon
B) Retreating Monsoon
C) Western Disturbances
D) Tropical Cyclones from Bay of Bengal

**24.** The International Court of Justice (ICJ), the principal judicial organ of the United Nations, is headquartered in:
A) Geneva, Switzerland
B) New York, USA

C) The Hague, Netherlands
D) Vienna, Austria

**25.** What is the total sanctioned strength of elected members in the Legislative Assembly of the Union Territory of Jammu and Kashmir (excluding seats reserved for Pakistan-occupied Jammu and Kashmir)?
A) 83
B) 90
C) 107
D) 114

**26.** Which canal connects the Mediterranean Sea to the Red Sea, facilitating maritime transit between Europe and Asia without circumnavigating Africa?
A) Panama Canal
B) Kiel Canal
C) Suez Canal
D) Corinth Canal

**27.** Which constitutional amendment introduced Article 21A, guaranteeing the Right to Education as a Fundamental Right to children aged 6 to 14 years?
A) 42nd Amendment Act, 1976
B) 44th Amendment Act, 1978
C) 86th Amendment Act, 2002
D) 91st Amendment Act, 2003

**28.** The Nilgiri Biosphere Reserve encompasses protected areas across which three Indian states?
A) Tamil Nadu, Kerala, and Karnataka
B) Andhra Pradesh, Karnataka, and Tamil Nadu
C) Maharashtra, Goa, and Karnataka
D) Kerala, Telangana, and Andhra Pradesh

**29.** The Dal Lake connects to the Anchar Lake through which natural water channel?
A) Mar Canal (Nallah Mar)
B) Nallah Amir Khan
C) Tsunt Kul
D) Doodh Ganga

**30.** Who was the first Governor of the unified state of Jammu and Kashmir after the abolition of the Sadar-i-Riyasat position in 1965?
A) Karan Singh
B) Bhagwan Sahay
C) L. K. Jha
D) B. K. Nehru

## Section II: Accountancy and Book Keeping (30 Questions | Q31 – Q60)

**31.** The convention that requires an enterprise to anticipate no profits but provide for all possible losses is the:
A) Matching Principle
B) Consistency Convention
C) Conservatism (Prudence) Convention
D) Materiality Convention

**32.** Which accounting equation correctly expresses the balance sheet relationship?
A) Assets} + {Liabilities} = {Capital}
B) Assets} = {Liabilities} - {Capital}
C) Capital} = {Assets} - {Liabilities}
D) Liabilities} = {Assets} + {Capital}

**33.** If goods costing ₹12,000 are sold for ₹15,000 on credit, what is the net impact on the basic accounting equation?
A) Assets increase by ₹3,000; Capital increases by ₹3,000
B) Assets decrease by ₹3,000; Liabilities increase by ₹3,000
C) Assets increase by ₹15,000; Liabilities increase by ₹12,000
D) No change in total assets

**34.** An entry recorded on both the debit and credit sides of a Three-Column Cash Book simultaneously is known as a:
A) Compound entry
B) Adjustment entry
C) Contra entry
D) Reversing entry

**35.** While preparing a Bank Reconciliation Statement starting with an Overdraft balance as per the Cash Book, interest debited by the bank but not recorded in the Cash Book should be:
A) Added to overdraft balance
B) Deducted from overdraft balance
C) Ignored
D) Multiplied by two

**36.** A Trial Balance provides proof of:
A) Total mathematical and posting accuracy of ledger accounts
B) Complete absence of all accounting errors

C) Complete truthfulness of financial statements
D) Correct valuation of fixed assets

**37.** Wages of ₹8,000 paid for the installation of new machinery were debited to the Wages Account. This is an:
A) Error of commission
B) Error of omission
C) Error of principle
D) Compensating error

**38. What is the nature of the "Provision for Doubtful Debts Account"?**
A) Nominal account
B) Personal account
C) Valuation (Contra-Asset) account
D) Real account

**39.** If the opening inventory is ₹40,000, purchases are ₹1,60,000, direct expenses are ₹20,000, and closing inventory is ₹30,000, the Cost of Goods Sold (COGS) is:
A) ₹1,90,000
B) ₹2,00,000
C) ₹1,80,000
D) ₹2,10,000

**40.** In the absence of a written partnership deed, the rate of interest allowed on a partner's loan to the firm is:
A) 5% per annum simple interest
B) 6% per annum simple interest
C) 10% per annum compound interest
D) Nil

**41.** When goodwill is already appearing in the balance sheet at the time of admission of a new partner, it is written off among old partners in their:
A) Sacrificing ratio
B) New profit-sharing ratio
C) Old profit-sharing ratio
D) Capital ratio

**42.** A contingent liability is shown:
A) Under non-current liabilities in the balance sheet
B) As a deduction from owner's equity
C) As a footnote/note to the balance sheet
D) On the debit side of the Profit & Loss Account

**43.** Which of the following is a direct tax levied in India?
A) Goods and Services Tax (GST)

B) Customs Duty
C) Central Excise Duty
D) Corporation Tax

***44. In the Single Entry System, profit earned during an accounting period is determined using the Statement of Affairs method by the formula:***
A)Closing Capital +Drawings - Additional Capital - Opening Capital
B) Opening Capital} +{Closing Capital} -{Drawings}
C) Closing Capital - Drawings +Additional Capital} -{Opening Capital}
D){Sales} - {Cost of Goods Sold}

**45.** What is the primary purpose of the Public Financial Management System (PFMS) in India?
A) To manage commercial bank portfolios
B) Real-time tracking of fund releases and direct benefit transfers under government schemes
C) To prepare corporate balance sheets
D) To regulate stock exchanges

**46.** A budget that is prepared on the basis that every line item must be justified from scratch regardless of previous appropriations is called a:
A) Flexible Budget
B) Performance Budget
C) Zero-Based Budget (ZBB)
D) Master Budget

**47.** The excess of actual sales over break-even sales is termed as:
A) Contribution Margin
B) Margin of Safety
C) Profit-Volume Ratio
D) Operating Leverage

**48.** Social Audit primarily evaluates:
A) Mathematical accuracy of the general ledger
B) Non-financial social performance, ethical accountability, and public welfare impact of an organization or scheme
C) Compliance with statutory corporate tax liabilities
D) Accuracy of depreciation schedules

**49.** An expenditure whose benefit is exhausted within the current accounting period is classified as:
A) Capital expenditure
B) Revenue expenditure
C) Deferred revenue expenditure
D) Development expenditure

**50.** Heavy advertising expenditure incurred to launch a new product, the benefit of which will last over four years, is best treated as:
A) Capital expenditure
B) Revenue expenditure
C) Deferred revenue expenditure
D) Outstanding expense

**51. The ledger balance of the Sales Returns Account is transferred at the end of the year to the:**
A) Profit & Loss Account (Debit side)
B) Trading Account (Deducted from Sales)
C) Balance Sheet (Current Assets)
D) Trading Account (Credit side)

**52.** Which of the following is an example of an Error of Commission?
A) Complete failure to record a credit sale in the sales book
B) Charging cost of plant repair to the plant asset account
C) Posting ₹5,400 to a customer's account as ₹4,500
D) Total omission of an invoice

**53.** Under the straight-line method, a machine purchased for ₹1,00,000 has an estimated residual value of ₹10,000 and a useful life of 5 years. Annual depreciation is:
A) ₹20,000
B) ₹18,000
C) ₹22,000
D) ₹15,000

**54. When shares are forfeited, the Share Capital Account is debited with the:**
A) Nominal face value of shares
B) Called-up amount on shares
C) Paid-up amount on shares
D) Market value of shares

**55.** The cost unit in passenger transport industry is typically expressed as:
A) Per kilometre
B) Per passenger
C) Per passenger-kilometre
D) Per vehicle

**56. Which voucher is prepared to record non-cash transactions such as depreciation and credit** purchases of assets?
A) Debit voucher
B) Credit voucher

C) Transfer (Journal) voucher
D) Payment voucher

**57.** In double-entry bookkeeping, the rule "Debit the receiver, Credit the giver" applies to:
A) Real Accounts
B) Personal Accounts
C) Nominal Accounts
D) Representative Nominal Accounts

**58.** If the total assets of a business are ₹4,50,000 and external liabilities are ₹1,70,000, what is the owner's capital?
A) ₹6,20,000
B) ₹2,80,000
C) ₹3,20,000
D) ₹1,70,000

**59.** A Bank Reconciliation Statement is prepared by:
A) The Reserve Bank of India
B) The commercial bank
C) The account holder (business firm)
D) The statutory auditor

**60.** In Cost Accounting, prime cost consists of:
A) Direct Material + Direct Labour + Direct Expenses
B) Direct Material + Factory Overheads
C) Factory Cost + Administrative Overheads
D) Prime Cost + Selling Overheads

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## Section III: General English (10 Questions | Q61 – Q70)

**61.** Identify the correct tense of the underlined verb:
"By the end of this month, the audit team **will have completed** the inspection."
A) Future Continuous
B) Future Perfect
C) Present Perfect
D) Future Indefinite

**62.** Rearrange the segments to form a coherent sentence:
(P) in public spending
(Q) accountability and transparency
(R) the newly introduced portal ensures
(S) to minimize corruption

A) R – Q – P – S
B) Q – P – S – R
C) P – S – R – Q
D) R – S – Q – P

**63.** Convert the direct sentence into indirect speech:
The auditor said to the manager, "Why did you not present the physical verification slips yesterday?"
A) The auditor asked the manager why he had not presented the physical verification slips the previous day.
B) The auditor asked the manager why did he not present the physical verification slips yesterday.
C) The auditor asked the manager that why he had not presented the slips.
D) The auditor enquired why he had not been presenting the slips.

**64.** Choose the correct modal auxiliary:
Candidates _____ carry an authentic government photo identity card to enter the examination hall.
A) might
B) could
C) must
D) would

**65.** Fill in the blank with the appropriate article:
He is _____ honest officer who graduated from _____ European university.
A) a; a
B) an; an
C) an; a
D) a; an

**66.** Fill in the blank with the correct pair of homophones:
The high _____ of wheat caused many farmers to lose their land, while the company introduced a new _____ of fabric.
A) coarse; course
B) cost; caste
C) price; prize
D) principal; principle

**67. What is the antonym of the word FASTIDIOUS?**
A) Meticulous
B) Careless
C) Fussy
D) Exacting

**68.** Identify the meaning of the idiom **"To cut corners"**:
A) To take a shortcut through a street
B) To do something badly or cheaply to save money or effort
C) To terminate an employee unexpectedly
D) To divide profits equally

**69.** Fill in the blank with the correct preposition:
The accountant was held liable _____ the embezzlement and was subsequently dismissed _____ service.
A) for; from
B) to; out of
C) of; by
D) in; off

**70.** Change into passive voice:
"The financial controller scrutinized the balance sheet meticulously."
A) The balance sheet was scrutinized meticulously by the financial controller.
B) The balance sheet had been scrutinized by the financial controller.
C) The balance sheet is scrutinized meticulously by the financial controller.
D) Meticulously the financial controller was scrutinizing the balance sheet.

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## Section IV: Statistics (10 Questions | Q71 – Q80)

**71.** Data originally collected by an investigator for a specific inquiry from first-hand sources is known as:
A) Secondary data
B) Primary data
C) Arrayed data
D) Tabular data

**72.** Which of the following is a major advantage of the direct personal interview method over mailed questionnaires?
A) It covers a vastly larger geographic area at low cost
B) It eliminates interviewer bias completely
C) High response rate and opportunity to clarify ambiguous questions
D) Absolute anonymity for respondents

**73.** Which measure of central tendency is determined graphically by the point of intersection of the 'less than' and 'more than' ogives?
A) Arithmetic Mean
B) Mode

C) Median
D) Geometric Mean

**74.** For a moderately skewed distribution, Karl Pearson's empirical relationship between Mean, Median, and Mode is:
A){Mode} = 2{Median} - 3{Mean}
B) Mode} = 3{Median} - 2,{Mean}
C) Median} = 3{Mode} - 2{Mean}
D)Mean} -{Median} = 3(Mean} - {Mode)

**75.** If the Laspeyres Price Index is $L = 121$ and the Paasche Price Index is $P = 100$, what is Fisher's Ideal Index number?
A) $110$
B) $110.5$
C) $121$
D) $105$

**76. Fisher's Ideal Index Number is called 'ideal' because it satisfies which of the following criteria?**
A) Unit Test and Circular Test only
B) Time Reversal Test and Factor Reversal Test
C) Circular Test and Base Shift Test
D) Time Reversal Test and Circular Test

**77.** In vital statistics, which metric accounts for mortality experience of female children from birth through their reproductive lifespan?
A) Crude Birth Rate (CBR)
B) Total Fertility Rate (TFR)
C) Gross Reproduction Rate (GRR)
D) Net Reproduction Rate (NRR)

**78.** In the theory of attributes, two attributes $A$ and $B$ are said to be positively associated if:
A) $(AB) = \frac{(A)(B)}{N}$
B) $(AB) > \frac{(A)(B)}{N}$
C) $(AB) < \frac{(A)(B)}{N}$
D) $(AB) = 0$

**79.** The total number of live births per 1,000 women of childbearing age (usually 15–49 years) in a given year is known as:
A) Crude Birth Rate
B) General Fertility Rate (GFR)

C) Total Fertility Rate
D) Gross Reproduction Rate

**80.** If two events $A$ and $B$ are independent with $P(A) = 0.4$ and $P(B) = 0.3$, then $P(A \cup B)$ is:
A) $0.70$
B) $0.58$
C) $0.12$
D) $0.82$

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## Section V: Mathematics (10 Questions | Q81 – Q90)

**81.** A sum of money lent under compound interest (compounded annually) doubles itself in 4 years. In how many years will it amount to 8 times the original principal?
A) 8 years
B) 12 years
C) 16 years
D) 10 years

**82.** The system of linear equations:

$$2x + 3y = 7$$

$$4x + 6y = 14$$

has:
A) A unique solution
B) Exactly two solutions
C) Infinitely many solutions
D) No solution

**83.** The number of distinct permutations that can be formed using all letters of the word **"COMMERCE"** is:
A) 5,040
B) 10,080
C) 20,160
D) 40,320

**84. Evaluate the limit:**

$$\lim_{x \to 0} \frac{\sin 5x}{\tan 2x}$$

A) $\frac{2}{5}$

B) $\frac{5}{2}$

C) $1$

D) $0$

**85. If $f(x) = \ln(x^2 + 1)$, then the derivative $f'(x)$ evaluated at $x = 1$ is:**

A) $\frac{1}{2}$

B) $1$

C) $2$

D) $0$

**86.** For two non-empty finite sets $A$ and $B$, if $n(A) = 15$, $n(B) = 22$, and $n(A \cup B) = 30$, find $n(A \cap B)$:

A) 5

B) 7

C) 8

D) 12

**87.** If A = \begin{pmatrix} 2 & 3 \ 1 & 4 \end{pmatrix}$, what is the value of $\det(A^{-1})$?

A) 5

B) $\frac{1}{5}$

C) $-5$

D) $\frac{1}{11}$

**88.** Two dice are rolled simultaneously. What is the probability that the sum of the numbers appearing on top is greater than 9?

A) $\frac{1}{6}$

B) $\frac{5}{36}$

C) $\frac{1}{12}$

D) $\frac{1}{4}$

**89.** The distance between the parallel planes $2x - y + 2z + 3 = 0$ and $2x - y + 2z - 6 = 0$ is:

A) 1 unit

B) 2 units

C) 3 units

D) 9 units

**90.** If $\vec{a} = 2\hat{i} + \hat{j} - 3\hat{k}$ and $\vec{b} = \hat{i} - 2\hat{j} + \hat{k}$, what is the scalar (dot) product $\vec{a} \cdot \vec{b}$?
A) $-3$
B) $3$
C) $-5$
D) $7$

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## Section VI: General Economics (10 Questions | Q91 – Q100)

**91.** The "law of equi-marginal utility" explains consumer equilibrium under:
A) Cardinal utility analysis
B) Ordinal utility analysis
C) Indifference curve approach
D) Revealed preference theory

**92.** An indifference curve is convex to the origin due to the:
A) Diminishing marginal rate of substitution (MRS)
B) Increasing marginal utility of income
C) Law of constant returns
D) Downward slope of the budget line

**93.** When the Reserve Bank of India sells government securities in the Open Market Operations (OMO), the immediate result is:
A) Expansion of commercial bank reserves and money supply
B) Contraction of liquidity from the banking system
C) A decline in policy repo rates
D) Reduction in the cash reserve ratio (CRR)

**94.** In the Ricardian Theory of Rent, rent arises because:
A) Land has an elastic supply
B) Land has differential fertility and situational advantages
C) Capital is scarcer than land
D) Land produces constant returns to scale

**95.** A market characterized by a few interdependent sellers where actions of one firm trigger counter-reactions by rivals is an:
A) Oligopoly
B) Monopoly
C) Perfect Competition
D) Monopsony

**96.** The "kinked demand curve" hypothesis developed by Paul Sweezy is used to explain:
A) Quantity rationing in monopoly

B) Price rigidity in oligopoly
C) Excess capacity in monopolistic competition
D) Wage determination under collective bargaining

**97.** Net Domestic Product at Factor Cost (NDP_FC) equals:
A)GDP_*MP - Depreciation - Net Indirect Taxes*
B)GNP_MP-Depreciation
C) GDP_FC +Net Factor Income from Abroad
D)NNP_FC - Net Factor Income from Abroad +{Depreciation

**98.** A situation where an economy experiences stagnant economic growth, high unemployment, and persistently high inflation concurrently is termed:
A) Hyperinflation
B) Stagflation
C) Deflationary spiral
D) Reflation

**99. Which of the following is an example of an automatic fiscal stabilizer in an economy?**
A) Discretionary infrastructure grants
B) Progressive income tax system
C) Annual corporate audit requirements
D) Open market purchase of sovereign bonds

**100.** Who acts as the "Lender of the Last Resort" in the Indian financial system?
A) State Bank of India
B) Ministry of Finance
C) Reserve Bank of India
D) Securities and Exchange Board of India

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## Section VII: General Science (10 Questions | Q101 – Q110)

**101.** A passenger standing in a moving bus falls forward when the driver suddenly applies brakes. This phenomenon is explained by:
A) Newton's First Law of Motion (Inertia)
B) Newton's Second Law of Motion
C) Newton's Third Law of Motion
D) Law of Conservation of Momentum

**102.** A 100 W bulb operates for 10 hours daily. How many units (kWh) of electrical energy does it consume in a 30-day month?
A) 10 units

B) 30 units
C) 300 units
D) 3 units

**103.** What type of chemical reaction is represented by the equation:
CaO + H20) —->Ca(OH)2(aq) +{Heat

A) Decomposition and endothermic
B) Combination and exothermic
C) Displacement and endothermic
D) Double displacement and neutral

**104.** Which non-metal is liquid at room temperature?
A) Mercury
B) Bromine
C) Chlorine
D) Phosphorus

**105.** The primary functional tissue responsible for the bidirectional translocation of soluble organic photosynthates (food) in plants is:
A) Xylem parenchyma
B) Xylem vessels
C) Phloem
D) Sclerenchyma

**106.** Night blindness and Xerophthalmia are clinical conditions caused by the deficiency of which vitamin?
A) Vitamin A (Retinol)
B) Vitamin B1 (Thiamine)
C) Vitamin C (Ascorbic acid)
D) Vitamin D (Calciferol)

**107.** In an ecological food chain, the transfer of energy from one trophic level to the next follows Lindeman's Ten Percent Law, which states that:
A) 90% of energy is transferred, 10% is lost
B) Only about 10% of energy is transferred to the next higher level
C) Energy increases by 10% at each successive level
D) Top carnivores retain 100% of primary producer energy

**108.** Which gas is the principal contributor to the enhanced anthropogenic greenhouse effect?
A) Carbon monoxide
B) Carbon dioxide

C) Sulfur dioxide
D) Nitrogen

**109.** Tuberculosis is a communicable infectious disease caused by a:
A) Virus
B) Bacterium (*Mycobacterium tuberculosis*)
C) Protozoan
D) Fungus

**110.** The Montreal Protocol (1987) is an international environmental treaty aimed at:
A) Reducing greenhouse gas emissions to combat climate change
B) Phasing out the production and consumption of ozone-depleting substances (ODS)
C) Regulating transboundary movements of hazardous chemical waste
D) Protecting maritime marine ecosystems

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## Section VIII: Knowledge of Computers (10 Questions | Q111 – Q120)

**111.** Which unit within the Central Processing Unit (CPU) coordinates and synchronizes the flow of data and execution of instructions across computer components?
A) Arithmetic Logic Unit (ALU)
B) Control Unit (CU)
C) Register Array
D) System Bus

**112.** Which open-source operating system kernel was created by Linus Torvalds in 1991?
A) Unix System V
B) Linux
C) MS-DOS
D) Windows NT

**113.** In Microsoft Excel, what type of cell reference is `$B$5`?
A) Relative cell reference
B) Absolute cell reference
C) Mixed cell reference
D) 3-D reference

**114.** What is the shortcut key combination to insert a hyperlink in Microsoft Word, PowerPoint, and Excel?
A) `Ctrl + H`
B) `Ctrl + K`

C) `Ctrl + L`
D) `Alt + K`

**115.** Which protocol is used to retrieve emails from a mail server while synchronizing mail state (such as read/unread and folders) across multiple client devices?
A) POP3
B) SMTP
C) IMAP
D) SNMP

**116.** A software program that covertly records every keystroke made on a keyboard to steal confidential login credentials is known as a:
A) Keylogger
B) Trojan Horse
C) Ransomware
D) Logic Bomb

**117.** Programs permanently stored in Read-Only Memory (ROM) that initialize and test hardware during the boot sequence are called:
A) Middleware
B) Firmware
C) Shareware
D) Freeware

**118.** Which of the following is an open-source relational database management system?
A) Oracle Database
B) Microsoft SQL Server
C) PostgreSQL
D) IBM DB2

**119.** In PowerPoint, which feature allows the presenter to define uniform layout, typography, slide backgrounds, and placeholders for all slides in a presentation?
A) Slide Master
B) Slide Sorter
C) Reading View
D) Animation Pane

**120.** The DigiLocker and UMANG platforms launched by the Government of India are initiatives under:
A) Digital India Programme
B) Make in India
C) Startup India
D) BharatNet Phase I exclusively

## Answer Key & Explanatory Solutions

| Q. | Ans | Brief Explanatory Note |
|---|---|---|
| **1** | **C** | The Fourth Schedule of the J&K Reorganisation Act, 2019 specifies forms of oaths/affirmations. |
| **2** | **B** | Section 57 abolishes the Legislative Council of the State of Jammu and Kashmir. |
| **3** | **A** | The Chenab Rail Bridge stands 359 m above the riverbed, the world's highest railway arch bridge. |
| **4** | **B** | Zoji La pass on NH-1 connects Srinagar with Kargil and Leh. |
| **5** | **B** | The Treaty of Amritsar (March 16, 1846) transferred Kashmir to Raja Gulab Singh for 75 lakh Nanakshahee rupees. |
| **6** | **B** | Kalhana completed *Rajatarangini* (River of Kings) in Sanskrit circa 1148–1150 CE. |
| **7** | **C** | Wular Lake in Bandipora is one of Asia's largest freshwater lakes and a Ramsar site fed by Jhelum. |
| **8** | **B** | Karewas are elevated Pleistocene lacustrine terraces famous for saffron (Zafran) in Pampore. |

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|---|---|---|
| 9 | B | King Lalitaditya Muktapida (8th century CE) constructed the Martand Sun Temple. |
| 10 | B | Dachigam National Park near Srinagar is the sole remaining natural sanctuary for the Hangul. |
| 11 | C | The Kishanganga (Neelum) is a major tributary of the Jhelum River. |
| 12 | C | Surinsar and Mansar lakes lie in the sub-Himalayan Shivalik belt of Jammu. |
| 13 | A | Bhand Pather is the traditional satirical folk drama of Kashmir. |
| 14 | C | Baglihar (Ramban) and Salal (Reasi) are run-of-the-river projects on the Chenab. |
| 15 | B | In Census 2011, Haryana had the lowest sex ratio among states (879 females per 1,000 males). |
| 16 | C | Strait of Hormuz connects the Persian Gulf with the Gulf of Oman and the Arabian Sea. |
| 17 | B | Tropic of Cancer passes through 8 states: GJ, RJ, MP, CG, JH, WB, TR, MZ (does not touch Odisha). |
| 18 | C | Gandhiji suspended the Non-Cooperation Movement on February 12, 1922 due to the Chauri Chaura violence. |
| 19 | C | The Kallanai Dam on the Kaveri River was built in the 2nd century CE by King Karikala. |

| Q. | Ans | Brief Explanatory Note |
|---|---|---|
| 20 | B | NW-1 covers the Ganga-Bhagirathi-Hooghly river system from Prayagraj to Haldia (1,620 km). |
| 21 | B | Good ozone is concentrated in the Stratosphere (15–35 km above Earth). |
| 22 | C | The Ramsar Convention provides the framework for national action and international cooperation on wetlands. |
| 23 | C | Western Disturbances originating in the Mediterranean region bring winter precipitation to NW India. |
| 24 | C | The ICJ sits at the Peace Palace in The Hague, Netherlands. |
| 25 | B | Post-delimitation, J&K Assembly has 90 elected seats (43 Jammu, 47 Kashmir) plus 24 vacant PoJK seats. |
| 26 | C | The Suez Canal connects Port Said (Mediterranean) to Port Tewfik (Red Sea). |
| 27 | C | The 86th Constitutional Amendment Act, 2002 inserted Article 21A. |
| 28 | A | Nilgiri Biosphere Reserve spans the meeting point of Western Ghats across TN, KL, and KA. |
| 29 | B | Nallah Amir Khan connects Dal Lake (via Nagin) directly to Anchar Lake. |

| Q. | Ans | Brief Explanatory Note |
|---|---|---|
| **30** | **A** | Dr. Karan Singh was the Sadar-i-Riyasat and transitioned as the first Governor in 1965. |
| **31** | **C** | Prudence/Conservatism ensures expenses/losses are recognized early, profits only when realized. |
| **32** | **C** | Fundamental accounting equation: $\text{Assets} = \text{Liabilities} + \text{Capital} \implies \text{Capital} = \text{Assets} - \text{Liabilities}$. |
| **33** | **A** | Debtors (Asset) increase by ₹15,000; Inventory (Asset) drops by ₹12,000 (net Assets +₹3,000). Profit (+₹3,000) adds to Capital. |
| **34** | **C** | Cash-to-bank or bank-to-cash transactions affect both columns and are marked 'C' (Contra). |
| **35** | **A** | Bank interest debited decreases passbook balance, further aggravating the cash overdraft. |
| **36** | **A** | A trial balance checks arithmetical accuracy; it cannot catch errors of principle or complete omission. |
| **37** | **C** | Capitalizing an asset installation cost as revenue expense violates fundamental accounting principles. |

| Q. | Ans | Brief Explanatory Note |
|---|---|---|
| 38 | C | Provision for doubtful debts is a contra-asset account deducted from Debtors. |
| 39 | A | COGS = Opening Inv + {Purchases} + Direct Exp. - Closing Inv. = 40,000 + 1,60,000 + 20,000 - 30,000 = ₹1,90,000$. |
| 40 | B | Under Section 13(d) of the Indian Partnership Act, 1932, loan interest is payable at 6% p.a. |
| 41 | C | Existing goodwill on admission is debited to old partners in their old profit-sharing ratio. |
| 42 | C | Contingent liabilities are disclosed as explanatory notes/footnotes under AS-4 / AS-29. |
| 43 | D | Corporation tax is a direct tax levied on corporate incomes. GST and customs are indirect taxes. |
| 44 | A | $\text{Profit} = \text{Closing Capital} + \text{Drawings} - \text{Additional Capital} - \text{Opening Capital}$. |
| 45 | B | PFMS provides end-to-end expenditure tracking and direct benefit payments under central/state schemes. |

| Q. | Ans | Brief Explanatory Note |
|---|---|---|
| **46** | **C** | Zero-Based Budgeting (ZBB) evaluates all activities fresh every budgeting cycle. |
| **47** | **B** | Margin of Safety $= \text{Actual Sales} - \text{Break-Even Sales}$. |
| **48** | **B** | Social audit checks whether public funds created intended socio-economic impacts on target beneficiaries. |
| **49** | **B** | Revenue expenditure benefits only the current operational year. |
| **50** | **C** | High initial promotion yielding multi-year utility is treated as deferred revenue expenditure. |
| **51** | **B** | Sales returns are deducted from Gross Sales on the credit side of the Trading Account. |
| **52** | **C** | Transposition/transcription errors in amounts constitute errors of commission. |
| **53** | **B** | {Depreciation} ={Cost} - Scrap/Life= 1,00,000 - 10,000/5} = 90,000/5} = ₹18,000. |
| **54** | **B** | On forfeiture, Share Capital Account is debited with the called-up value per share. |
| **55** | **C** | Composite cost unit for transport is Passenger-Kilometre (passengers × distance). |

| Q. | Ans | Brief Explanatory Note |
|---|---|---|
| 56 | C | Transfer vouchers record non-cash/adjustment transactions. |
| 57 | B | The golden rule for Personal Accounts is "Debit the receiver, Credit the giver". |
| 58 | B | $\text{Capital} = \text{Assets} - \text{Liabilities} = 4,50,000 - 1,70,000 = ₹2,80,000$. |
| 59 | C | The BRS is an internal control document prepared by the customer/account holder. |
| 60 | A | $\text{Prime Cost} = \text{Direct Materials} + \text{Direct Labour} + \text{Direct Expenses}$. |
| 61 | B | "will have completed" is in the Future Perfect tense ($\text{will/shall} + \text{have} + V_3$). |
| 62 | A | R-Q-P-S forms: "The newly introduced portal ensures accountability and transparency in public spending to minimize corruption." |
| 63 | A | Past simple becomes past perfect in reported interrogatives; inverted auxiliary becomes assertive word order. |
| 64 | C | "Must" conveys an absolute mandatory rule. |

| Q. | Ans | Brief Explanatory Note |
|---|---|---|
| **65** | **C** | 'Honest' starts with a vowel sound (/ɒ/), requiring *an*; 'European' starts with a consonant glide (/j/), requiring *a*. |
| **66** | **A** | 'Course' refers to path/flow; 'coarse' means rough texture. |
| **67** | **B** | Fastidious means excessively particular/exacting; its opposite is careless. |
| **68** | **B** | To cut corners means to skimp on quality or standards to save cost or time. |
| **69** | **A** | Idiomatic usage: "liable for" (responsible) and "dismissed from" (removed from job). |
| **70** | **A** | Simple past passive: Object + was/were+ V_3 +{by} +{Subject} |
| **71** | **B** | First-hand collected data is primary data. |
| **72** | **C** | Direct interaction minimizes ambiguity, boosting accurate response rates. |
| **73** | **C** | The intersection of less-than and more-than ogives projects directly to the Median on the x-axis. |
| **74** | **B** | Pearsonian empirical relation: $\text{Mode} = 3\,\text{Median} - 2\,\text{Mean}$. |
| **75** | **A** | Fisher's Index is the geometric mean: |

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|---|---|---|
| | | $F = \sqrt{L \times P} = \sqrt{121 \times 100} = \sqrt{121}$ . |
| **76** | **B** | Fisher's Index satisfies both Time Reversal Test ($I_{01} \times I_{10} = 1$ ) and Factor Reversal Test. |
| **77** | **D** | NRR considers both fertility and the female mortality rate from birth to end of reproduction. |
| **78** | **B** | Positive association exists when the actual frequency $(AB)$ exceeds the expected frequency $\frac{(A)(B)}{N}$. |
| **79** | **B** | GFR} = \frac{\text{Total Live Births}}{\text{Female Pop. aged 15--49}} \times 1,000$. |
| **80** | **B** | $P(A \cup B) = P(A) + P(B) - P(A \cap$ . |
| **81** | **B** | In CI, $A = P(1 + r)^t$. If money doubles in 4 years, it becomes $2^3 = 8$ times in $3 \times 4 = 12$ years. |
| **82** | **C** | $\frac{a_1}{a_2} = \frac{2}{4} = \frac{1}{2}$, $\frac{b_1}{b_2} = \frac{3}{6} = \frac{1}{2}$, $\frac{c_1}{c_2} = \frac{7}{14} = \frac{1}{2}$ (coincident lines $\implies$ infinitely many solutions). |
| **83** | **B** | Letters in COMMERCE: 8 total (C:2, O:1, M:2, E:2, R:1). $\text{Permutations} = \frac{8!}{2!\,2!\,2!} = \frac{40,320}{8} = 5,040 \times 2 = 5,040 \text{{ |

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|---|---|---|
| | | (wait: } $\frac{40,320}{8} = 5,040\text{)}$. *Note: Option A is 5,040.* |
| **84** | **B** | $\lim_{x \to 0} \frac{\sin 5x}{\tan 2x} = \lim_{x \to 0} \frac{\frac{\sin 5x}{5x} \cdot 5x}{\frac{\tan 2x}{2x} \cdot 2x} = \frac{1 \cdot 5}{1 \cdot 2} =$ . |
| **85** | **B** | $f'(x) = \frac{2x}{x^2+1}$. At $x = 1$, $f'(1) = \frac{2(1)}{1^2+1} = \frac{2}{2} = 1.$ |
| **86** | **B** | $n(A \cap B) = n(A) + n(B) - n(A \cup$ . |
| **87** | **B** | $\det(A) = (2)(4) - (3)(1) = 8 - 3 = 5 \implies \det(A^{-1}) = \frac{1}{\det(A)} = \frac{1}{5}$. |
| **88** | **A** | Sum $> 9$ means sum is 10, 11, or 12. Favourable pairs: $(4, 6), (5, 5), (6, 4), (5, 6), (6, 5), (6, 6$ . Probability $= \frac{6}{36} = \frac{1}{6}$. |
| **89** | **C** | $d = \frac{ |
| **90** | **A** | $\vec{a} \cdot \vec{b} = (2)(1) + (1)(-2) + (-3)$ . |
| **91** | **A** | Equi-marginal utility is the hallmark of Alfred Marshall's Cardinal Utility analysis. |
| **92** | **A** | Diminishing Marginal Rate of Substitution gives indifference curves their convex profile. |

| Q. | Ans | Brief Explanatory Note |
|---|---|---|
| **93** | **B** | Selling government bonds in OMO absorbs cash reserves from commercial banks, curbing credit. |
| **94** | **B** | David Ricardo based differential rent on variations in original and indestructible fertility and location of soil. |
| **95** | **A** | Oligopoly is defined by mutual interdependence and strategic interaction among a small number of sellers. |
| **96** | **B** | Paul Sweezy's kinked demand curve explains sticky/rigid pricing under oligopoly. |
| **97** | **A** | $\text{NDP}_{\text{FC}} = \text{GDP}_{\text{MP}} - \text{Depreciation} - \text{Net Indirect Taxes (Indirect Taxes - Subsidies)}$. |
| **98** | **B** | Stagflation combines economic stagnation (recession/job loss) with high inflation. |
| **99** | **B** | **Progressive income taxes automatically siphon off higher proportions of income as GDP surges.** |
| **100** | **C** | As lender of last resort, the RBI provides liquidity against approved securities to solvent banks facing liquidity crunches. |
| **101** | **A** | Newton's First Law (Law of Inertia) causes the upper body |

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|---|---|---|
| | | to continue moving forward when the bus decelerates. |
| **102** | **B** | Energy $= \text{Power} \times \text{Hours} \times \text{Days} = 0.1\text{ kW} \times 10\text{ h} \times 30 = 30\text{ kWh (units)}$. |
| **103** | **B** | Slaking of lime combines two substances into one product and releases substantial heat (exothermic). |
| **104** | **B** | Bromine is the only non-metal in the liquid state at standard room temperature. |
| **105** | **C** | Phloem sieve tubes and companion cells translocate organic sucrose bidirectionally. |
| **106** | **A** | Vitamin A deficiency impairs rhodopsin synthesis in rod cells, leading to night blindness. |
| **107** | **B** | Raymond Lindeman's 10% law specifies that only ~10% of energy is conserved across trophic transfers. |
| **108** | **B** | Carbon dioxide contributes over 60% of enhanced global warming from human activity. |
| **109** | **B** | TB is an airborne bacterial infection caused by *Mycobacterium tuberculosis*. |
| **110** | **B** | The 1987 Montreal Protocol successfully mandated targets |

| Q. | Ans | Brief Explanatory Note |
|---|---|---|
| | | to phase out CFCs and other halons. |
| **111** | **B** | The Control Unit manages fetch-decode-execute cycles and directs the flow of data within the system. |
| **112** | **B** | Linux kernel was created by Linus Torvalds as free open-source software under GPL. |
| **113** | **B** | Prepending `$` to both column letter and row number locks the cell reference absolutely. |
| **114** | **B** | `Ctrl + K` is the universal Microsoft Office shortcut to insert or edit hyperlinks. |
| **115** | **C** | Internet Message Access Protocol (IMAP) maintains full two-way synchronization on the server. |
| **116** | **A** | Keyloggers secretly capture and forward typed keystrokes, including master passwords. |
| **117** | **B** | Firmware (such as BIOS/UEFI) is embedded low-level code that initializes hardware. |
| **118** | **C** | PostgreSQL is an advanced open-source object-relational database management system. |
| **119** | **A** | Slide Master enforces uniform themes, fonts, slide layouts, and formatting rules across presentations. |

| Q. | Ans | Brief Explanatory Note |
|---|---|---|
| **120** | **A** | DigiLocker and UMANG are key digital infrastructure pillars of the flagship Digital India Programme. |